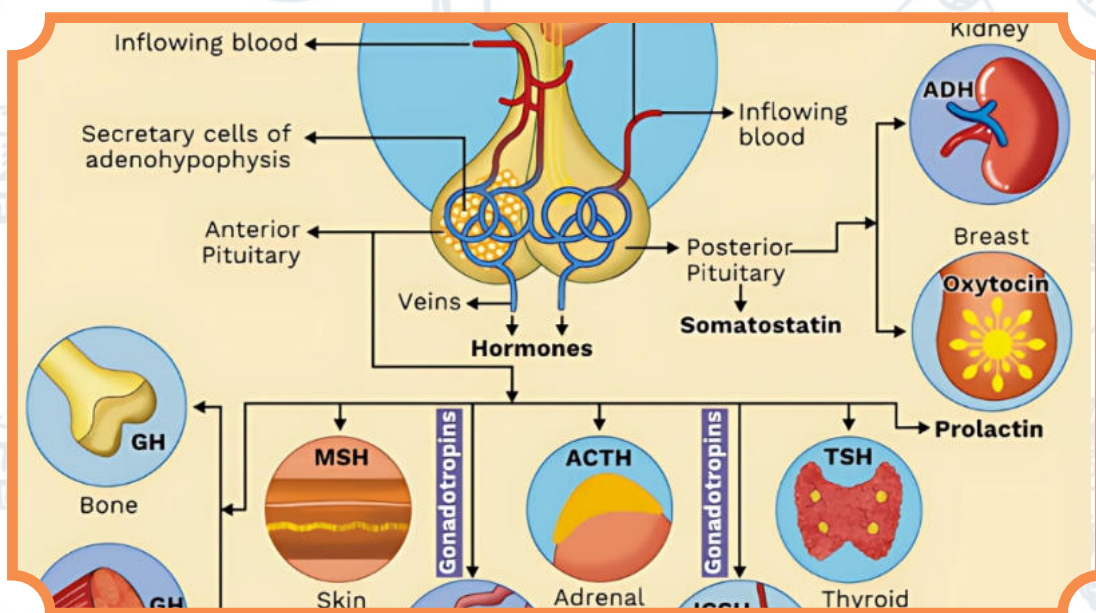


UNIT

18

CHEMICAL COORDINATION





Background:

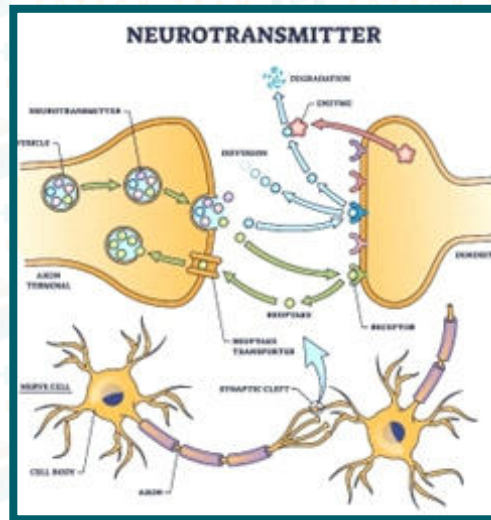
- As **complex multicellular organisms** evolved, they developed the need to **coordinate** activities between cells in different parts of the body.
- Cell-to-cell communication** is essential for controlling:
 - Movement
 - Growth
 - Reproduction
 - Maintenance of homeostasis (stable internal environment)

Mechanisms of Cell Communication:

Different mechanisms have evolved to allow cells within organisms to communicate:

1. Neurotransmitters:

- Chemical messengers between neurons (nerve cells).
- Already discussed in the previous chapter.



2. Pheromones:

- Chemicals secreted by an organism in minute amounts.
- Stimulate specific reactions in another organism of the same species.
- Example:** Used in mating or marking territory.





3. Hormones:

- (Not detailed in your text but important to include)
- Chemical messengers **secreted by endocrine glands**.
- Transported through the **bloodstream** to target organs or tissues.
- Help regulate processes like growth, metabolism, reproduction, and stress response.



Importance of Chemical Coordination:

- Ensures that **different parts** of the body work together smoothly.
- Allows organisms to **respond to internal and external changes** efficiently.
- Maintains **balance** and **stability** in biological systems (homeostasis).

CHEMICAL COORDINATION — HORMONES:

Definition of Hormone:

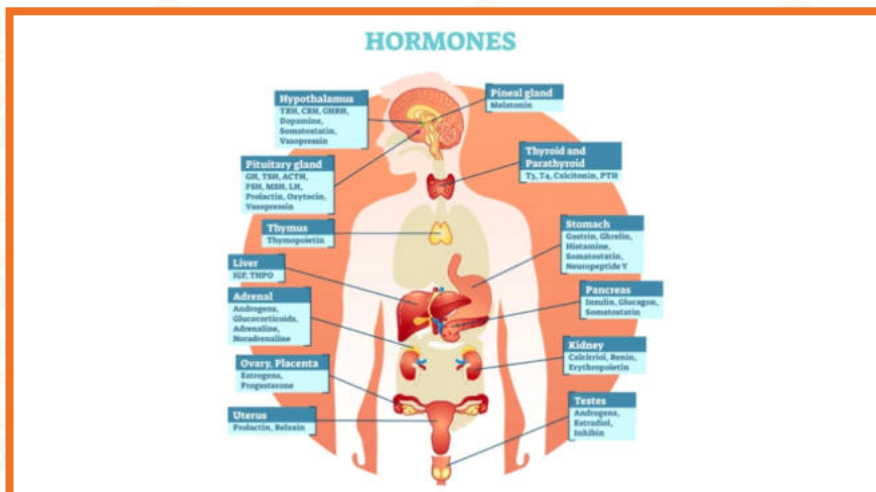
- (Greek: Hormon = excite)
- A **hormone** is a **chemical messenger** secreted by specialized tissues called **glands**.
- Transported through the **bloodstream** to reach various parts of the body.
- Only **specific target cells** (with the right receptors) can respond to a given hormone — other cells **ignore it**.

CHEMICAL NATURE OF HORMONES:

Basic Characteristics of Hormones:

To function effectively, hormones must:

1. Be **sufficiently complex** to convey regulatory information to target cells.
2. Be **adequately stable** to avoid destruction before reaching the target.

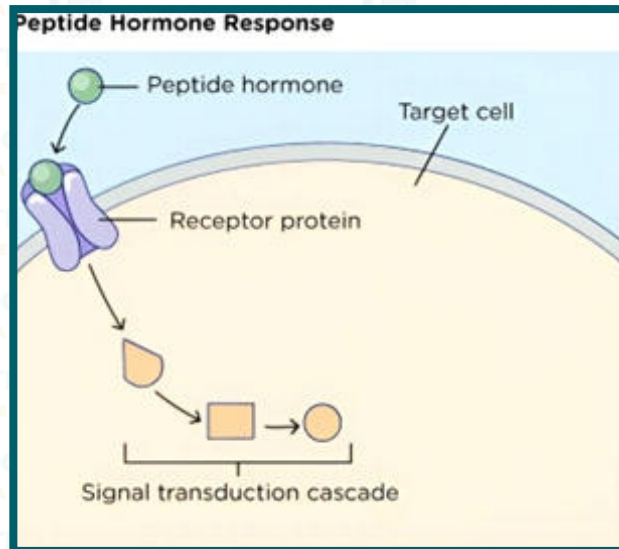




Three Primary Chemical Categories of Hormones:

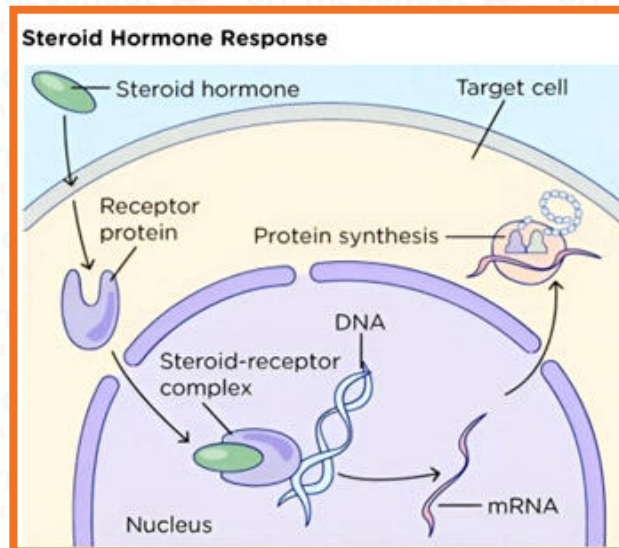
1. Peptide and Protein Hormones:

- Made of **amino acid chains**.
- **Peptide hormones:** Short chains (e.g., ADH = Antidiuretic hormone).
- **Protein hormones:** Long chains (e.g., GH = Growth hormone, Insulin).



2. Steroid Hormones:

- Made by **enzymatic modification** of cholesterol.
- Examples: Testosterone, Estrogen (Estradiol), Progesterone, Aldosterone, Cortisol.



3. Amino Acid Derivative Hormones:

- Made by **enzymatic modification** of specific amino acids.
- Includes **biogenic amines** (nervous coordination):
- **Adrenal medulla:** Catecholamines (from tyrosine) → epinephrine (adrenaline), norepinephrine (noradrenaline).
- **Thyroid hormones:** Also derived from tyrosine.
- **Pineal gland:** Melatonin (derived from tryptophan).





Path of Chemical Messenger

Hormone Transport and Targeting

- **All cells** have blood supply, so hormones in blood can reach almost every cell.
- However, hormones only affect **target cells** that have **specific receptors** for that hormone.
- If a cell **lacks the receptor**, it will **not** respond to the hormone.

Hormone Specificity

- The **same hormone** can cause **different effects** depending on the type of target cell.

Receptor Locations on Target Cells

- Found in **two general locations**:
 1. **On the cell membrane** (surface receptors).
 2. **Inside the cell** (intracellular receptors).

MODE OF HORMONE ACTION

General Mechanisms

- Hormones trigger changes in target cells by two general mechanisms depending on their **chemical nature**:
 - **Hydrophilic (polar)** → protein/peptide hormones
 - **Lipophilic (non-polar)** → steroid and thyroid hormones

1. Protein Hormones (Hydrophilic Hormones)

- **Nature**:
 - Water-soluble
 - Lipid-insoluble → **cannot cross cell membrane**
- **Action Mechanism**:
 - Bind to **protein receptors** on the **outer surface** of the target cell membrane.
 - Often trigger **rapid, short-term responses**.

Second Messenger System

- Frequently used by protein hormones.
- **Steps**:
 1. Hormone (primary messenger) binds to surface receptor.
 2. Receptor's shape changes → triggers a cascade of biochemical reactions.
 3. Often, this activates an enzyme that converts **ATP** into **cyclic AMP (cAMP)**.
 4. **cAMP** acts as a **second messenger** → relays the signal inside the cell.

Effects of Hormone Signal (via cAMP)

- Induction of **protein** synthesis.
- Activation or inactivation of **enzymes**.
- Change in **cell membrane permeability**.
- Altering **mitosis** and **cell development rates**.
- Induction of **product secretion**.

2. Steroid Hormones (Lipophilic Hormones)

Nature and Properties

- Includes steroid hormones and thyroid hormones.
- Lipid-soluble → can easily diffuse through the cell membrane.
- Unlike protein hormones, they do not need membrane receptors.





Mechanism of Action

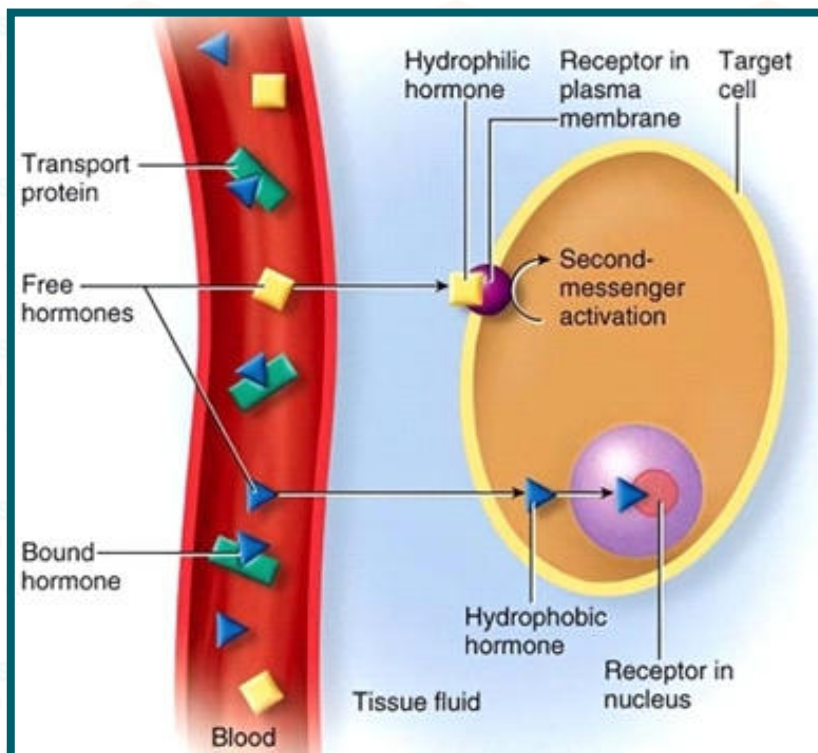
1. Hormones **enter the target cell** by diffusing through the lipid bilayer.
2. Inside the cell, they **bind to specific protein receptors** — often located in the **nucleus**.
3. The **hormone-receptor complex** binds to **DNA** at specific sites.
4. This triggers **transcription**: messenger RNA (mRNA) is synthesized from specific genes.
5. The **mRNA** moves to the **cytoplasm** and directs the **synthesis of new proteins**.

Effect Timeline

- **Slower response** compared to protein hormones.
- Effects may take **minutes to days** to become fully visible.
- Leads to **long-term changes** in cell function by altering **gene expression**.

Summary of Key Effects

- Regulation of **gene activity**.
- Production of **new proteins** that change cell structure or function.
- Long-lasting impacts on processes like development, metabolism, and reproduction.





Comparison Table: Hormone Modes of Action

Feature	Protein (Hydrophilic) Hormones	Steroid (Lipophilic) Hormones
Examples	Insulin, ADH, Growth Hormone, Catecholamines	Testosterone, Estrogen, Progesterone, Cortisol
Chemical Nature	Water-soluble, lipid-insoluble	Lipid-soluble, water-insoluble
Receptor Location	On cell membrane surface	Inside the cell (cytoplasm or nucleus)
Ability to cross membrane	Cannot cross lipid bilayer	Can easily cross lipid bilayer
Signal pathway	Uses second messenger system (e.g., cAMP)	Directly interacts with DNA or nuclear machinery
Primary messenger	Hormone binds to external receptor → activates enzymes	Hormone passes through membrane → binds internal receptor
Speed of action	Fast, short-term effects	Slower, long-term effects
Typical effects	Enzyme activation/inactivation, membrane changes, secretion	Gene expression changes, protein synthesis, long-term cell function changes

Flowchart: Protein Hormone Action (via cAMP)

- 1. Hormone binds to membrane receptor
- 2. Receptor shape changes → activates G-protein or enzyme
- 3. ATP → cAMP (cyclic AMP, second messenger)
- 4. cAMP activates intracellular pathways
- 5. Final cellular response:
 - Enzyme activity changes
 - Membrane permeability changes
 - Protein synthesis induction
 - Secretion or metabolic changes

ENDOCRINE SYSTEM OF MAN

Definition

- The endocrine system consists of all the glands and tissues that secrete hormones.
- Examples of endocrine glands:
 - Pituitary gland
 - Thyroid gland
 - Adrenal gland





How Endocrine Glands Work

- Cells in endocrine glands secrete hormones into **extracellular fluid** → hormones **diffuse into blood capillaries**.
- Because hormones enter the bloodstream, they are called **endocrine secretions**.

Difference Between Endocrine and Exocrine Glands

Endocrine Glands	Exocrine Glands
Secrete hormones into blood (no ducts)	Secrete products into ducts or onto body surfaces (e.g., gut, skin)
Example: Pituitary, thyroid, adrenal glands	Example: Pancreas (secretes digestive enzymes into small intestine)

Hypothalamus

Location & Role

- Part of the **brain** containing specialized cells called **neurosecretory cells**.
- Main role: **regulate the pituitary gland** (also known as the “master gland”).

Functions

1. Control of Anterior Pituitary

- Neurosecretory cells secrete two types of hormones into the blood:
- Releasing hormones** → stimulate the anterior pituitary to secrete hormones.
- Inhibiting hormones** → stop the anterior pituitary from secreting hormones.

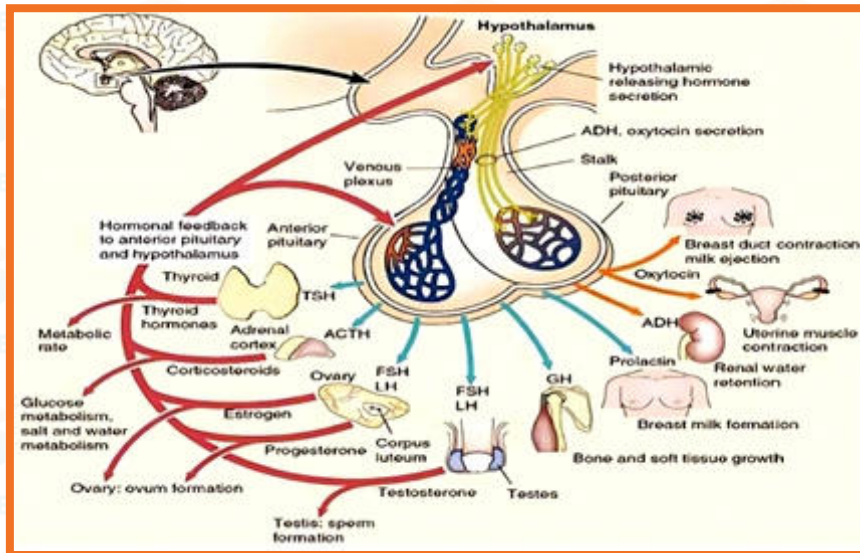
2. Production of Hormones Stored in Posterior Pituitary

- Another set of neurosecretory cells in the hypothalamus produces:
- ADH (Antidiuretic Hormone)**
- Oxytocin**
- These are stored and secreted from the posterior pituitary.

Hypothalamic Hormones and Their Effects

Hormone from Hypothalamus	Effect on Anterior Pituitary
Thyrotropin Releasing Hormone (TRH)	Stimulates release of TSH (Thyroid Stimulating Hormone)
Corticotropin Releasing Hormone (CRH)	Stimulates release of ACTH (Adrenocorticotrophic Hormone)
Gonadotropin Releasing Hormone (GnRH)	Stimulates release of LH and FSH
Growth Hormone Releasing Hormone (GHRH)	Stimulates release of GH (Growth Hormone)
Growth Hormone Inhibiting Hormone (GHIH)	Inhibits secretion of GH
Prolactin Inhibiting Hormone (PIH)	Inhibits secretion of Prolactin
Melanocyte Inhibiting Hormone (MIH)	Inhibits secretion of MSH (Melanocyte Stimulating Hormone)





PITUITARY GLAND

Overview

- Also known as **hypophysis**.
- Pea-sized gland** hanging from the hypothalamus by a stalk.
- Divided into **two distinct lobes**:
- 1. **Anterior Pituitary (Adenohypophysis)** → glandular part.
- 2. **Posterior Pituitary (Neurohypophysis)** → fibrous part.

Anterior Pituitary (Adenohypophysis)

- Functions as an **independent endocrine gland**.
- Produces **at least seven essential hormones**.
- Many are **tropic hormones (tropins)** → hormones that stimulate other endocrine glands.

Major Hormones of Anterior Pituitary

Hormone	Function
Adrenocorticotrophic Hormone (ACTH)	Stimulates adrenal cortex → produces corticosteroids.
Melanocyte Stimulating Hormone (MSH)	Stimulates synthesis and dispersion of melanin pigment in the skin.
Growth Hormone (GH) / Somatotrophic Hormone (STH)	Stimulates growth of muscles, bones, tissues.
Prolactin	Stimulates milk production in mammary glands.
Thyroid Stimulating Hormone (TSH)	Stimulates thyroid gland → produces thyroxin.
Luteinizing Hormone (LH)	Stimulates gonads:





- Ovaries → estrogen, progesterone
 - Testes → testosterone
 - Follicle Stimulating Hormone (FSH)
 - Females → development of ovarian follicles
 - Males → required for sperm development
- (*FSH + LH are collectively called gonadotropins)

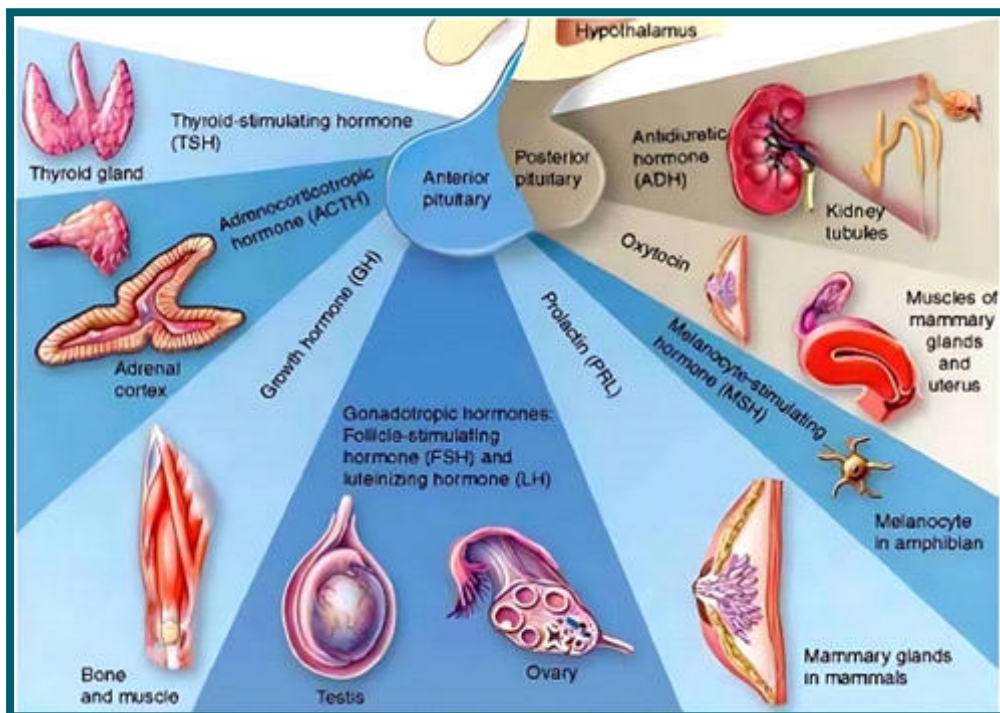
Posterior Pituitary (Neurohypophysis)

- Contains **nerve endings** of neurosecretory cells whose bodies are in the hypothalamus.
- Does **not** produce hormones itself — only **stores and releases** hormones made by the hypothalamus.

Hormone	Function
Antidiuretic Hormone (ADH)	Prevents urination by increasing water reabsorption in the kidneys (collecting ducts), helping maintain body water balance and prevent dehydration.

Oxytocin

- Stimulates **uterine contractions** during labor.
- Promotes **milk ejection** (lactation) after childbirth.
- Acts as a **chemical messenger** in the brain, influencing **social behavior** and bonding.





THYROID GLAND

Location & Structure

- Located at the base of the neck, in front of the trachea (windpipe).
- Bi-lobed (two lobes) connected by a narrow band called the isthmus.

Major Hormones Produced

Hormone	Details & Function
Tri-iodothyronine (T ₃)	Contains 3 iodine atoms ; more potent but secreted in smaller amounts.
Tetra-iodothyronine (T ₄)/Thyroxin	Contains 4 iodine atoms ; secreted in larger amounts but less potent than T ₃ .
Calcitonin	Peptide hormone; helps regulate calcium (Ca ²⁺) levels in the blood.

Functions of Thyroid Hormones (T₃ & T₄)

- Elevate metabolic rate in most body cells.
- Increase oxygen consumption and heart rate.
- Stimulate synthesis of enzymes involved in glucose breakdown for energy.
- Help regulate body temperature, especially under cold conditions.

Hormonal Regulation (Feedback Loop)

- Hypothalamus secretes TRH (Thyrotropin-Releasing Hormone) ☒
- Stimulates Anterior Pituitary to release TSH (Thyroid Stimulating Hormone) ☒
- Stimulates Thyroid Gland to release T₃ and T₄.

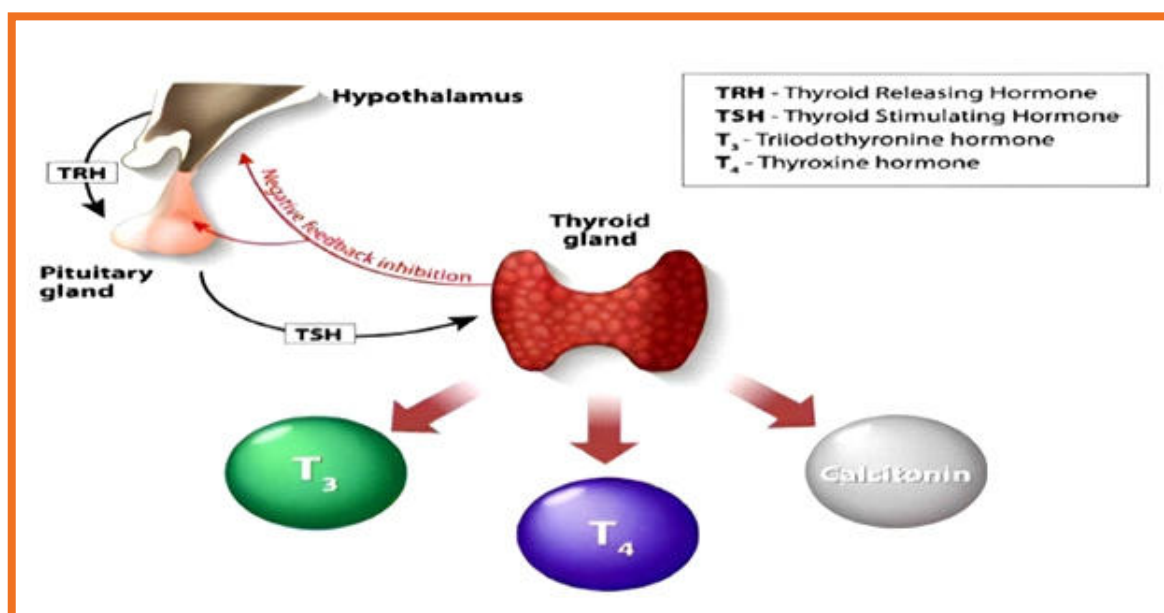
Disorders Related to Thyroid Function

Condition	Cause	Symptoms
Hyperthyroidism	Excessive thyroid hormone secretion	Profuse sweating, weight loss, heat intolerance, irritability, high blood pressure.
Hypothyroidism	Low thyroid hormone secretion	Weight gain, lethargy, cold intolerance (in adults).
Cretinism	Congenital hypothyroidism in children	Stunted growth, severe intellectual disability.
Goiter	Thyroid gland enlargement due to iodine deficiency	Enlarged thyroid, often due to iodine deficiency in diet.

Function of Calcitonin

- Helps maintain calcium balance in the blood.
- When blood Ca²⁺ levels are too high → calcitonin stimulates calcium uptake into bones, lowering blood calcium concentration.





PARATHYROID GLAND

Location & Structure

- **Four small glands** located on the **back of the thyroid gland**. **Hormone Produced**
- **Parathormone (PTH)** or **Parathyroid Hormone**

A **peptide hormone**.

Secreted in response to **falling calcium (Ca^{2+}) levels** in the blood.

Functions of Parathyroid Hormone (PTH)

Target	Action
Bones	Stimulates osteoclasts to break down bone matrix (calcium phosphate crystals) → releases calcium into blood.
Kidneys	Stimulates reabsorption of calcium from urine → conserves calcium in the body.
Intestine (indirectly)	Activates Vitamin D, which enhances calcium absorption from food in the intestine.

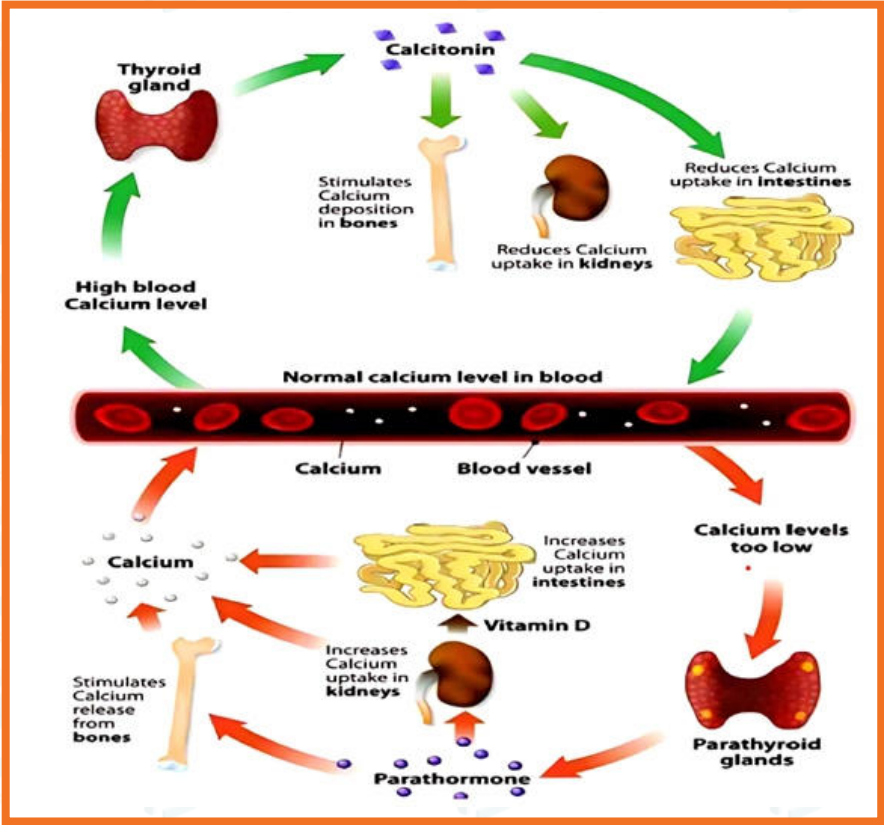
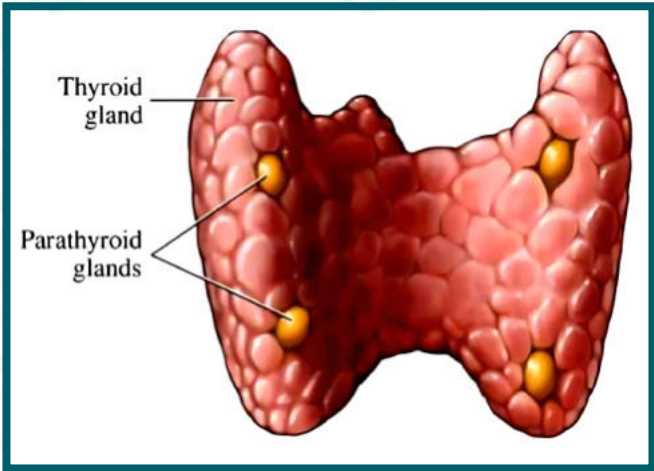
Importance of Blood Calcium Regulation

- Essential for muscle function (including the heart).
- Vital for nervous system and endocrine system functions.
- A significant drop in blood calcium can lead to severe muscle spasms and serious physiological disturbances.

Feature	Calcitonin	Parathyroid Hormone (PTH)
Source	Thyroid gland (parafollicular / C cells)	Parathyroid glands (small glands behind thyroid)
Type of hormone	Peptide hormone	Peptide hormone
Stimulus for release	High blood calcium ($\uparrow \text{Ca}^{2+}$ levels)	Low blood calcium ($\downarrow \text{Ca}^{2+}$ levels)
Main effect	Lowers blood calcium levels	Raises blood calcium levels



Feature	Calcitonin	Parathyroid Hormone (PTH)
Action on bones	Inhibits osteoclast activity → decreases bone resorption → calcium stored in bone	Stimulates osteoclast activity → increases bone resorption → calcium released into blood
Action on kidneys	Promotes calcium excretion in urine	Promotes calcium reabsorption in kidneys
Action on intestines	No direct effect	Activates Vitamin D → increases calcium absorption from intestine
Overall role	Prevents excess calcium in blood (calcium homeostasis)	Maintains minimum calcium level in blood (calcium homeostasis)





PANCREAS

Location & Structure

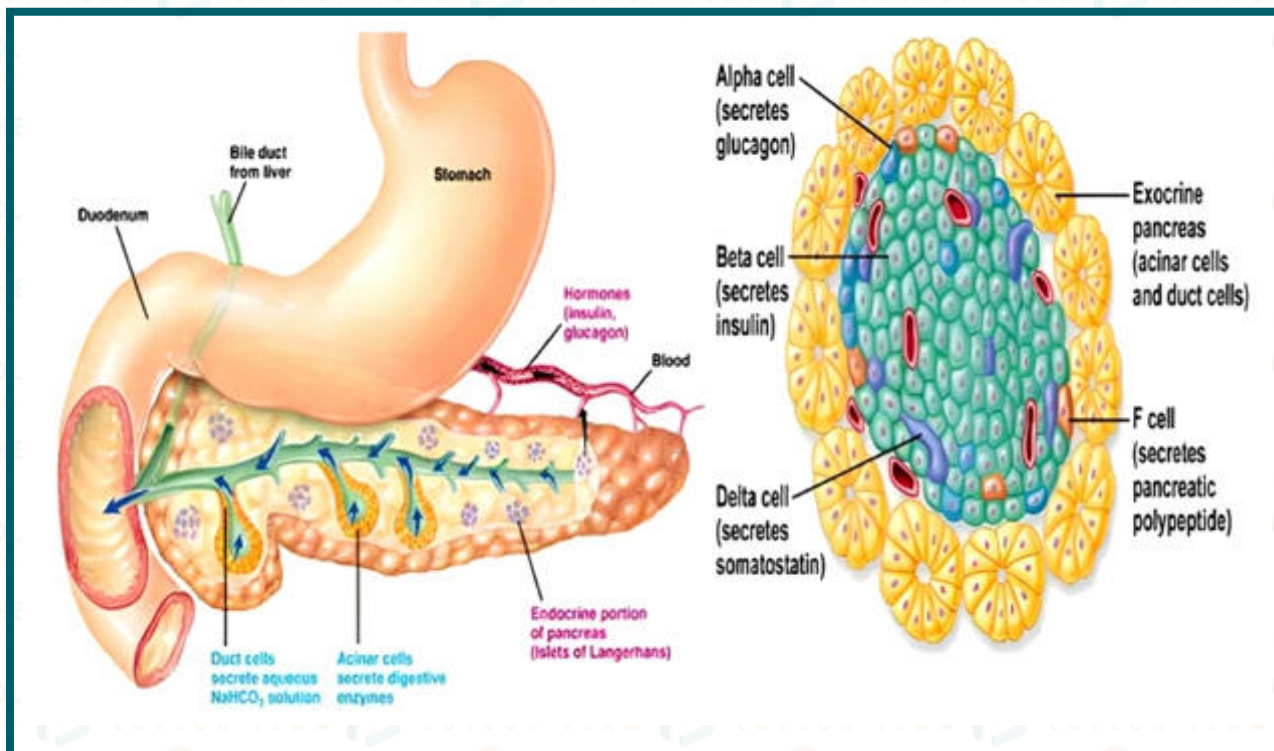
- Located adjacent to the stomach.
- Dual function: Both endocrine (hormonal) and exocrine (digestive) functions.
- Endocrine cells make up only 2% of the pancreas, the remaining 98% consists of exocrine tissues.
- Exocrine tissues secrete bicarbonate ions and digestive enzymes, which are transported to the small intestine via the pancreatic duct.

Endocrine Function (Islets of Langerhans)

- In 1869, German student Paul Langerhans discovered clusters of endocrine cells called Islets of Langerhans in the pancreas.
- These endocrine cells secrete two important hormones into the bloodstream:
- Alpha cells: Secrete glucagon (peptide hormone).
- Beta cells: Secrete insulin (peptide hormone).

Hormones of the Pancreas

Hormone	Secreted by	Function	Action
Insulin	Beta cells of Islets of Langerhans	Lowers blood glucose levels (after eating).	Stimulates cells to take up glucose for energy or convert it to fat/glycogen for storage. Major storage organ is the liver
Glucagon	Alpha cells of Islets of Langerhans	Raises blood glucose levels (when blood glucose drops).	Stimulates the liver to break down glycogen into glucose and release it into the bloodstream. Also promotes breakdown of fats to release fatty acids.





Regulation of Blood Glucose Levels

High blood glucose (after eating):

- **Insulin** is secreted promotes glucose uptake by cells & lowers blood glucose levels.

Low blood glucose (e.g., skipping meals):

- **Glucagon** is secreted → stimulates glycogen breakdown in liver & raises blood glucose levels.

Key Points

- Insulin and glucagon are **antagonistic hormones**:
- **Insulin** lowers blood glucose.
- **Glucagon** raises blood glucose.
- **Liver** is the main organ for storing **glycogen**.

DIABETES & EFFECTS OF LACK OF INSULIN

Effects of Insulin Deficiency

- **Lack of functional** insulin in diabetics leads to reliance on **fats** as the primary energy source.
- This results in **high circulating levels of lipids** (including **cholesterol**).
- **Severe diabetes** can cause:
- **Fatty deposits** in blood vessels → increases **high blood pressure** and contributes to **heart disease**.
- **Fat deposits in small blood vessels** can damage the **retina**, leading to **blindness**.
- Deposits in the **kidneys** can cause **kidney failure**.

TYPES OF DIABETES

Diabetes Type-I (Insulin-Dependent Diabetes)

- **Cause: Autoimmune disorder** → immune system attacks and destroys the **beta (β) cells** in the **Islets of Langerhans**.
- Result: **Little or no insulin** is produced.
- **Symptoms:** High blood glucose levels due to lack of insulin.
- **Treatment:** Managed by **insulin injections** to control blood sugar levels.

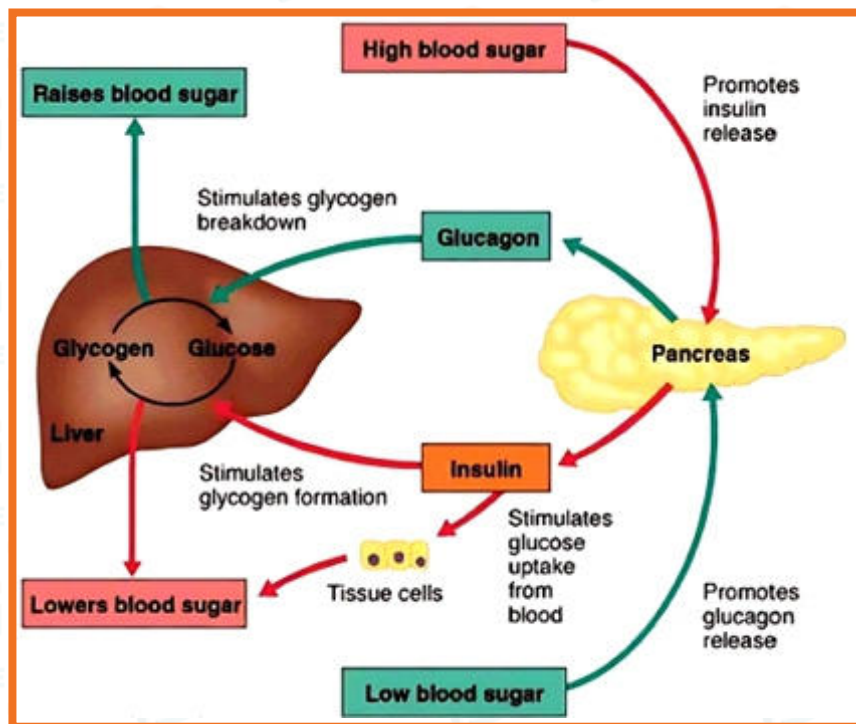
Diabetes Type-II (Non-Insulin-Dependent Diabetes)

- **Cause:** Associated with **genetic factors, obesity, stress, lack of exercise, and old age**.
- **Insulin production** is normal, but **target cells** do not respond effectively to insulin (insulin resistance).
- **Symptoms:** Insulin is present, but the body is unable to use it efficiently, leading to high blood glucose levels.
- **Management:** Can be controlled by adopting a **low carbohydrate diet**, increasing physical activity, and managing stress.

Key Points to Remember

- **Type-I Diabetes:** Autoimmune destruction of **beta cells** → no insulin production.
- **Type-II Diabetes:** Insulin resistance at the **target cells** → insulin is produced, but cells don't respond.

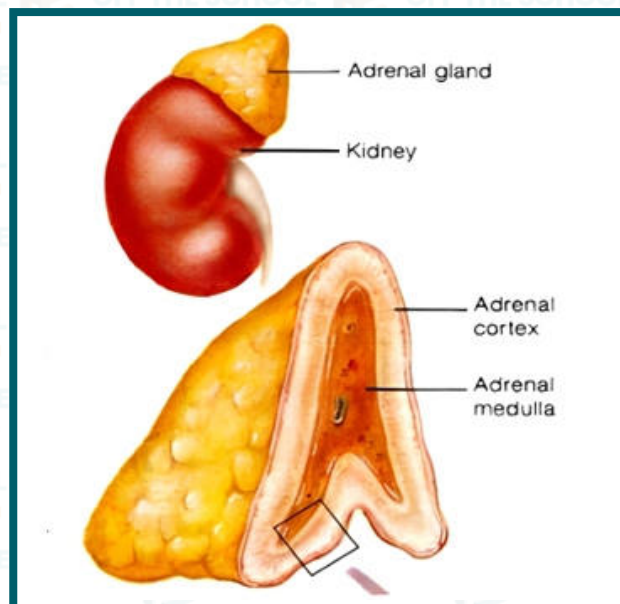




ADRENAL GLAND

Location & Structure

- Located **above each kidney**.
- Composed of two main parts:
 - **Adrenal Medulla** (inner portion).
 - **Adrenal Cortex** (outer portion).



Adrenal Medulla (Central Part)

- **Function:** Produces two key hormones in response to **stress**:
 - **Epinephrine** (adrenaline)
 - **Norepinephrine** (nor-adrenaline)
- **Role:** These hormones trigger "**alarm**" responses similar to those of the sympathetic nervous system, preparing the body for **extreme efforts** (fight or flight response).





Effects of Epinephrine and Norepinephrine:

- **Increased heart rate**
- **Increased blood pressure**
- **Dilation of bronchioles** (better airflow to lungs)
- **Elevated blood glucose** (providing energy)
- **Reduced blood flow to skin and digestive organs**
- **Increased blood flow to heart and muscles**
- **Activation:** The adrenal medulla is directly activated by the **sympathetic nervous system** during emergencies.

Adrenal Cortex (Outer Portion)

- **Function:** Secretes **steroid hormones**, collectively called **corticosteroids**.
- The three main types of corticosteroids are:
 1. **Glucocorticoids** (e.g., **cortisol**)
 2. **Mineralocorticoids** (e.g., **aldosterone**)
 3. **Androgens** (male sex hormones)

Glucocorticoids (e.g., Cortisol):

- **Function:** Stimulate the **breakdown of muscle protein** into amino acids, which are transported by the blood to the liver for **gluconeogenesis** (production of glucose).
- **Effect:** Helps the body respond to stress by increasing energy availability.

ADRENAL GLAND

Glucocorticoids (e.g., Cortisol)

- **Function:**
 - Stimulate the **liver** to produce enzymes for **gluconeogenesis** (conversion of amino acids into glucose).
 - Increase **glucose synthesis** and encourage the use of **lipids** rather than glucose for energy.
- **Regulation:**
 - **ACTH (Adrenocorticotrophic Hormone)** from the anterior pituitary stimulates glucocorticoid release.
 - **Corticotropin-Releasing Hormone (CRH)** from the hypothalamus stimulates ACTH release in response to stress (e.g., trauma, infection, severe temperatures).
- **Metabolic Role:**
 - Glucocorticoids act similarly to **glucagon**, maintaining blood glucose levels, especially for the brain, which requires glucose for energy.
- **Effects:**
 - **High cortisol levels** can lead to **Cushing's syndrome** (Hyper-cortisolism) with symptoms like:
 - High blood pressure
 - Weight gain
 - Muscle wasting
 - Weakened bones
 - Mood swings
 - **Low cortisol levels** lead to **Addison's disease**, which is characterized by:
 - Insufficient energy supply
 - Low blood glucose
 - High susceptibility to stress
 - Risk of death from even mild infections due to the body's inability to cope with stress.





Mineralocorticoids (e.g., Aldosterone)

- **Function:**
 - Regulate mineral balance, particularly the reabsorption of **sodium (Na⁺)** in the kidneys.
- **Aldosterone** helps maintain **blood volume** and **blood pressure** by preventing sodium loss from the urine.
- **Regulation:**
 - When **blood sodium levels** are low, the kidneys secrete an enzyme called **renin**, which activates a cascade known as the **Renin-Angiotensin-Aldosterone System (RAAS)**:
 - Renin converts **angiotensinogen** into **angiotensin-I**, which is then converted to **angiotensin-II** in the lungs.
 - Angiotensin-II stimulates the release of **aldosterone** from the adrenal cortex.
- **Effects:**
 - **Angiotensin-II** constricts **arteries** to increase blood pressure.
 - **Aldosterone** increases sodium absorption in the kidneys, which increases **blood volume** and **blood pressure**.

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM (RAAS)

- **Function:**

The **RAAS** is a hormone system that regulates **blood pressure** and **fluid balance** by controlling **sodium** and **water** retention in the kidneys.
- **Process:**
 - **Renin** is secreted by the **juxtaglomerular cells** of the kidneys when **blood pressure** or **sodium levels** are low.
 - Renin converts a plasma protein called **angiotensinogen** (produced by the liver) into **angiotensin I**.
 - **Angiotensin I** is converted to **angiotensin II** by an enzyme called **angiotensin-converting enzyme (ACE)** in the lungs.
 - **Angiotensin II** has two primary effects:
 1. It causes **vasoconstriction**, narrowing blood vessels and increasing **blood pressure**.
 2. It stimulates the release of **aldosterone** from the adrenal cortex.
- **Aldosterone's Role:**
 - **Aldosterone** increases **sodium reabsorption** by the kidneys, leading to **water retention** and an increase in **blood volume** and **blood pressure**.

● **Outcome:**
The overall effect of the RAAS system is to increase **blood pressure** and **blood volume**, ensuring proper kidney function and fluid balance.

CORTICOSTEROIDS: ANDROGENS

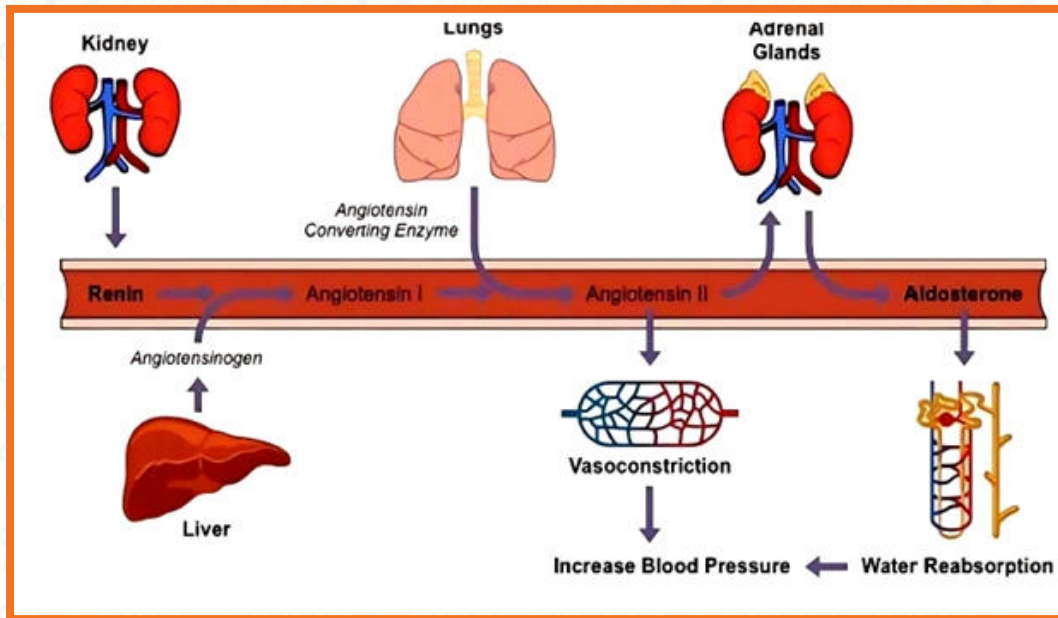
- **Type:**

One of the groups of hormones produced by the adrenal cortex is **sex hormones**, specifically **androgens**, which are similar to **testosterone**.
- **Function:**
 - **Androgens** play a role in the development of **secondary sexual characteristics** during **adolescence**. These changes occur in both males and females, though the effects are more pronounced in males.





- In males, androgens promote the development of features such as facial hair, deepening of the voice, and increased muscle mass.
- In females, androgens contribute to the development of pubic and underarm hair and influence libido.
- **Presence:**
 - Androgens are produced in both males and females, but they are secreted in larger amounts in males during puberty.



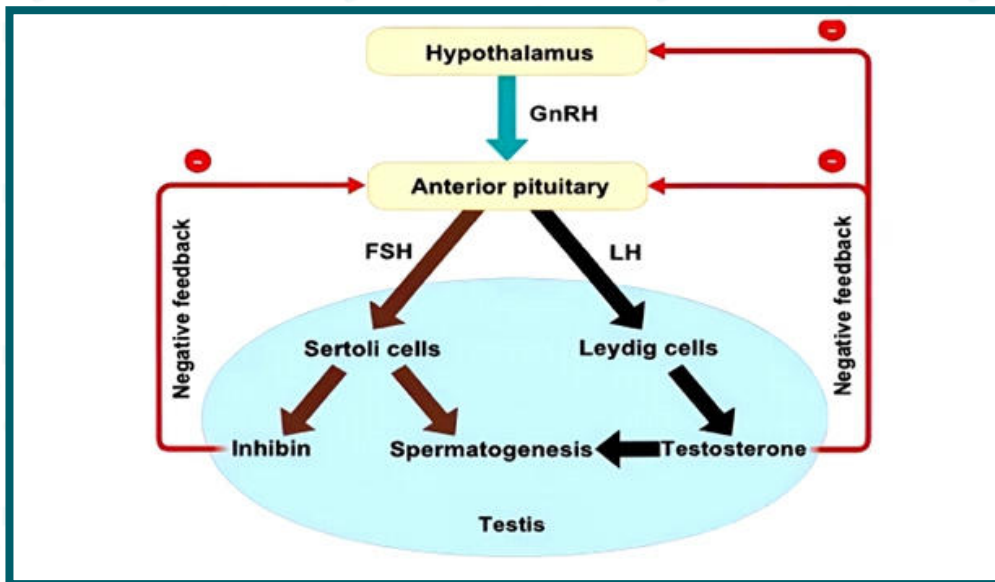
GONADS

The gonads are the primary reproductive organs that produce **sex hormones**. These hormones play a critical role in sexual development, reproductive functions, and secondary sex characteristics.

Testes (Male Gonads)

- **Functions:**
 - Produce **sperm** and **male sex hormones** (primarily testosterone).
 - Testosterone is secreted by **Leydig cells** (located between seminiferous tubules).
 - The anterior pituitary gland releases **gonadotropins: Follicle Stimulating Hormone (FSH)** and **Luteinizing Hormone (LH)**.
- **FSH** stimulates **Sertoli cells** in the testes for sperm development.
- **LH** stimulates **Leydig cells** to secrete testosterone.
- **Testosterone Effects:**
 - **Anabolic effects** (growth-promoting):
 - Increases muscle **mass, strength, bone density**, and **linear growth**.
 - **Androgenic effects** (male characteristics):
 - Maturation of sex organs, **deepening of the voice**, **formation of scrotum** in fetus, and growth of **facial and axillary hairs**.
- **Development of male sex organs** is determined early in embryo development by testosterone.





Ovaries (Female Gonads)

- **Functions:**
- Produce **eggs (ova)** and **female sex hormones: estrogen and progesterone.**
- **Estrogen:**
- Promotes the development of **female reproductive organs.**
- Contributes to **secondary sexual characteristics:**
- Development of **breasts, fat distribution** (hips, legs, breasts), and **pubic** and **armpit hairs.**
- **Menarche** (start of the menstrual cycle).
- Regulates **body temperature**, improves **blood flow**, and has protective effects against certain diseases like **colon cancer, Parkinson's disease, and Alzheimer's disease.**
- **Progesterone:**
- Regulates **menstruation** and **maintains pregnancy.**
- **Estrogen's 400 functions:**
- **Mood regulation, decreases wrinkling**, and supports overall **health.**

Other Endocrine Tissues

Several tissues and organs outside the primary endocrine glands also produce hormones:

Gastrin (Stomach):

- Enhances **gastric motility**, promotes the secretion of **hydrochloric acid (HCL)**, and supports **mucosal growth.**

Secretin (Duodenum):

- Regulates the pH environment of the stomach, pancreas, and liver to create a neutral/basic pH in the duodenum.

Cholecystikinin (Duodenum):

- Stimulates the gallbladder to contract and promotes the release of digestive enzymes from the pancreas.

Kidney:

- **Angiotensin:** Regulates blood pressure.
- **Erythropoietin:** Stimulates the production of red blood cells in bone marrow.





Heart:

- **Atrial Natriuretic Hormone:** Increases **salt and water excretion** to help reduce **blood pressure**.

Prostaglandins:

- **Lipophilic hormones** involved in **inflammation** and **blood flow**, also playing a role during **pregnancy**.

Placenta:

- Secretes **human chorionic gonadotropin (hCG)** and **progesterone** during pregnancy.

Brain:

- Releases **endorphins**, which help relieve pain and contribute to **emotional well-being**.

FEEDBACK MECHANISM

Feedback mechanisms are essential for regulating most of the body's functions. They work similarly to a thermostat, responding to changes in the system by either increasing or decreasing the activity of certain processes to maintain homeostasis (balance).

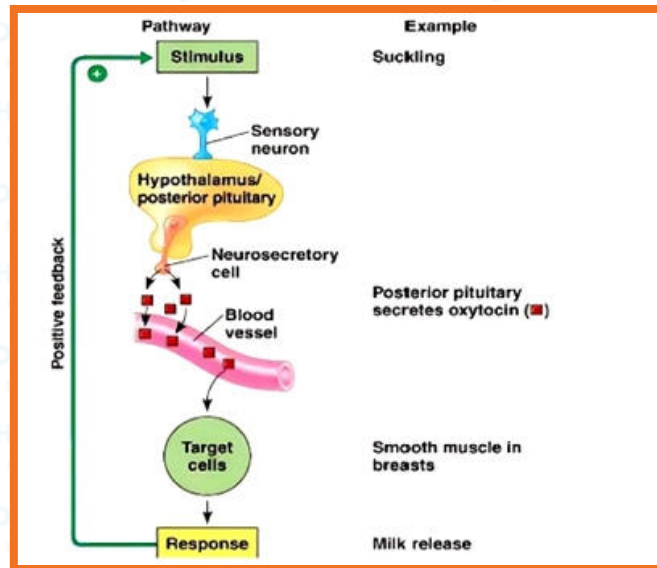
TYPES OF FEEDBACK MECHANISMS

There are two main types of feedback mechanisms: Positive Feedback and Negative Feedback.

1. Positive Feedback Mechanism

- **Definition:** In a positive feedback mechanism, the release of a hormone leads to an action that causes even more of the hormone to be released, enhancing the initial response. This cycle continues until a specific endpoint is reached.
 - **Characteristics:**
 - Positive feedback is rare in the endocrine system.
 - It amplifies or accelerates a particular action.
 - **Examples:**
 - **Childbirth:**
 - During labor, the release of **oxytocin** stimulates uterine contractions.
 - These contractions cause more oxytocin to be released, which leads to stronger contractions.
 - This cycle continues until the baby is born, at which point the feedback loop ends.
 - **Breastfeeding:**
 - When the baby sucks on the nipple, sensory neurons send signals to the **hypothalamus** and **posterior pituitary**.
 - This stimulates the release of **oxytocin**, which causes **milk ejection** from the breasts.
 - The milk release further stimulates more oxytocin release, continuing the cycle until milk is ejected.
 - **Illustration:**
 - **Childbirth (Oxytocin):** The release of oxytocin leads to uterine contractions, which causes more oxytocin to be released, intensifying contractions.





2. Negative Feedback Mechanism

- **Definition:** Negative feedback mechanisms are more common in the endocrine system. They work to maintain homeostasis by counteracting changes. When a hormone level is too high or too low, the system responds by either decreasing or increasing the production of that hormone to bring it back to normal.

- **Characteristics:**

- Negative feedback helps the body maintain **stability**.
- It works to **counteract** any deviations from a set point (homeostasis).

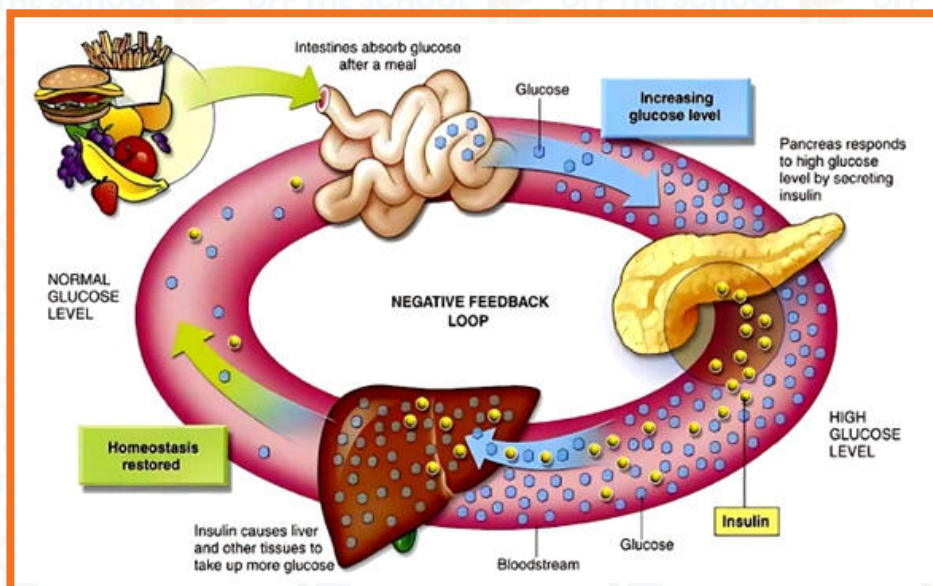
- **Example:**

- **Blood Glucose Regulation:**

- When blood glucose levels rise (after eating), the pancreas releases **insulin**.
- **Insulin** helps cells absorb glucose, lowering blood sugar levels back to normal.
- Once glucose levels return to normal, the pancreas stops releasing insulin, maintaining balance.

- **Illustration:**

- Blood Glucose Regulation: The pancreas detects rising blood glucose, releases insulin to lower glucose levels, and stops releasing insulin once normal levels are reached.





Summary:

- **Positive Feedback:** Amplifies or accelerates a process (e.g., childbirth, breastfeeding).
 - **Negative Feedback:** Counteracts changes to maintain stability (e.g., blood glucose regulation).
- These feedback mechanisms help maintain the balance of hormones and other bodily functions to ensure proper physiological conditions.

